INCH-POUND
MS24571J
7 August 2014
SUPERSEDING
MS24571H
21 April 2009

MILITARY SPECIFICATION SHEET
CIRCUIT BREAKERS, AIRCRAFT, TRIP FREE, 2 ½ TO 50 AMPERES, 121.1°C AMBIENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27715.

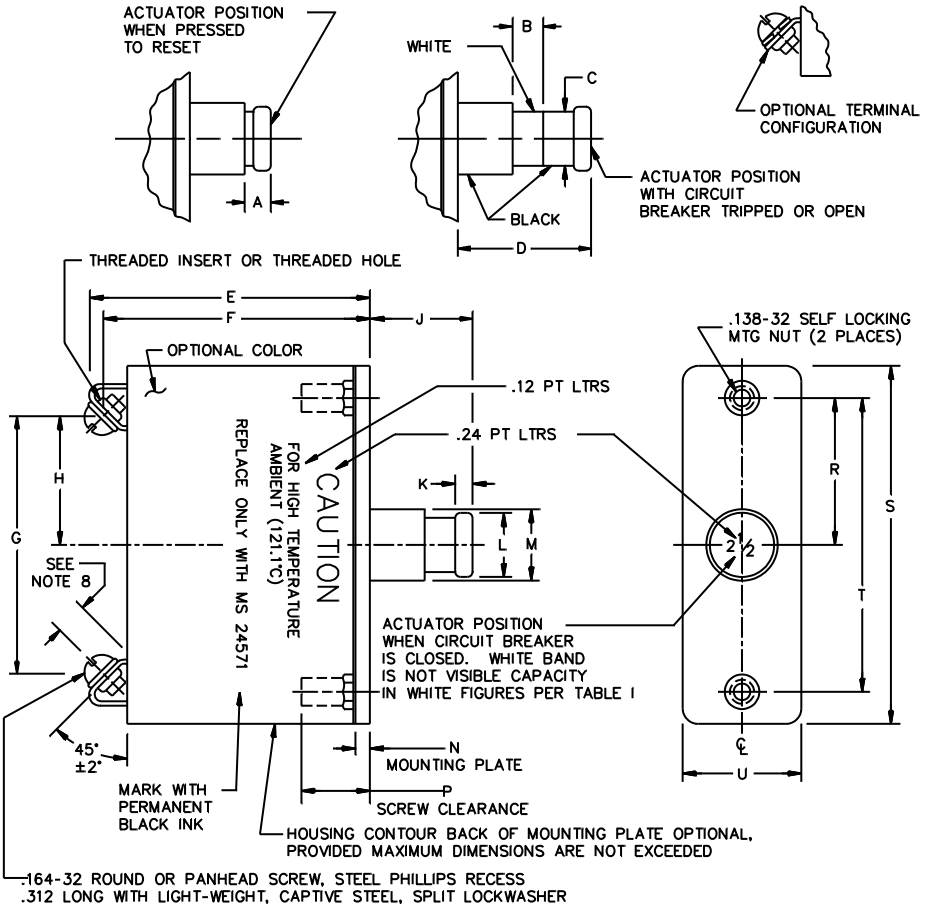


FIGURE 1. Dimensions and configuration.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	.120	---	3.048	---
B	.187	.249	4.75	6.32
C	.340	.370	8.64	9.40
D	.690	.750	17.526	19.050
E	2.340	2.400	59.36	60.960
F	2.150	2.210	54.610	56.134
G	1.160	1.220	29.464	30.988
H	.564	.624	14.326	15.850
J	.500	.532	12.70	13.51
K	.060	.120	1.524	3.048
L	.402	.432	10.21	10.97
M	.453	.483	11.51	12.27
N	.031	---	.78	---
P	.312	---	7.92	---
R	.900	.912	22.86	23.16
S	---	2.189	---	55.60
T	1.800	1.820	45.72	46.23
U	---	.760	---	19.30

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .030 (0.762 mm). Angles ±2°.
4. In the event of a conflict between the text of this standard and the references sited herein, the text of this standard shall take precedence.
5. Referenced Government documents listed in the Department of Defense ASSIST Online database form a part of this standard to the extent specified herein.
6. Part or identifying number:

MS24571 - 2V

MS number — dash number (see table I and note 7.)

7. For dash numbers ending with “V”, the vibration test methods shall be as follows:
   Operating: Method 204, Condition C of MIL-STD-202.
   Nonoperating: Method 204, Condition B of MIL-STD-202.
8. .250 Min. Typ. clearance from center line of screw to case.
9. The terminals shall not be moveable with respect to case.
10. The terminals shall not be identified as to “line” or “load”.

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Electrical and mechanical characteristics.

Dash number		Capacity amperes	Voltage drop Max	Weight Max lbs	Operating force lbs				Endurance cycles		
					Pullout		Reset		Resistive 1/	Inductive 1/	Mech no load
					Max	Min	Max	Min			
-2	-2V	2 ½	0.6	0.2	8	1.5	12	4	5,000 0.95 ±.05 PF	5,000 0.75 Max Lag PF	10,000
-5	-5V	5	0.4	0.2	8	1.5	12	4			
-7	-7V	7 ½	0.3	0.2	8	1.5	12	4			
-10	-10V	10	0.25	0.2	8	1.5	12	4			
-15	-15V	15	0.25	0.2	8	1.5	12	4			
-20	-20V	20	0.25	0.2	8	1.5	12	4			
-25	-25V	25	0.25	0.2	8	1.5	12	4			
-35	-35V	35	0.25	0.2	8	1.5	12	4			
-50	-50V-	50	0.25	0.2	8	1.5	12	4			

1/ 400 Hz 115/200 volt system, tested at 120 ±5 volts 380 - 420 Hz.

Table II Detail calibration requirements.

Dash number		Tripping time from -53.9°C to +93.3°C (time in seconds)							
		Percent rated current							
		200		400		1000		3000	
		Min	Max	Min	Max	Min	Max	Min	Max
-2	-2V	12	32	2.3	6.5	.32	1.1	.04	.155
-5	-5V	14	36	2.5	6.5	.45	1.1	.06	.13
-7	-7V	16	40	2.5	7.0	.34	1.1	.04	.12
-10	-10V	18	42	2.7	7.75	.45	1.2	.05	.125
-15	-15V	20	45	2.7	8.0	.36	1.2	.04	.13
-20	-20V	20	48	2.7	8.0	.36	1.2	.04	.12
-25	-25V	20	50	2.7	9.5	.36	1.2	.04	.12
-35	-35V	20	53	2.7	8.0	.36	1.2	.04	.12
-50	-50V	20	55	2.7	10.0	.36	1.4	---	---

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TABLE III. Detail rupture current performance.

Test Condition	System	Voltage before fault	Calibrated fault current (amperes)	Transient voltage after fault current interruption	Open circuit voltage
L-N	400 Hz 115/200 volts	120 ± 5	6000 amperes during first cycle of fault current decreasing to 3600 amperes within .05 second after initiation		120 ± 5
L-L	400 Hz 115/200 volts	200 ± 8	4200 amperes during first cycle of fault current decreasing to 2500 amperes within .05 second after initiation		200 ± 8
	28 volts dc	30 ± 2	6000 amperes in .01 to .03 second after initiation	28 volts within .002 second after interruption 50 volts max.	30 ± 2

TABLE IV. Coordination test pairs.

Lower rating (amperes)	Higher rating (amperes)
2 ½	5
5	10
7 ½	15
10	20
15	35

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TABLE V. Coordination trip time.

Rating (amperes)	Trip time (seconds) at 2000 percent rated current min.
2 ½	
5	.124
7 ½	.088
10	.112
15	.089
20	.103
25	.102
35	.109
50	---

Nominal voltage rating:
28 Vdc or
115 V line to neutral or
200 V line to 400 Hz

Referenced documents. In addition to MIL-DTL-27715, this document references the following:

MIL-STD-202

The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5925-2014-010)

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